

Theorem 1 (Fundamental Theorem of Arithmetic). *Every integer greater than 1 has a unique factorization into primes.*

Lemma 1 (Pigeonhole). *If $n + 1$ items are placed in n boxes, some box contains at least two items.*

Corollary 1 (Infinitely Many Primes). *There is no largest prime.*

Proposition 1 (Even Squares). *If n is even, then n^2 is divisible by 4.*

Conjecture 1 (Twin Prime Conjecture). *There are infinitely many primes p such that $p + 2$ is also prime.*

Algorithm 1 (Euclid's Algorithm). *To compute $\gcd(a, b)$, replace (a, b) with $(b, a \bmod b)$ until $b = 0$; the remaining a is the $\gcd$.*

Definition 1 (Prime). *A positive integer $p > 1$ is prime if its only positive divisors are 1 and p .*

Example 1 (Composite). $6 = 2 \times 3$, so 6 is not prime.

Exercise 1 (Factorize). *Find the prime factorization of 360.*

Proof 1 (Proof). Write $n = 2k$; then $n^2 = 4k^2$.

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Remark 1 (Remark). The infinitude of primes was proved by Euclid around 300 BCE.

Solution 1 (Solution). $360 = 2^3 \times 3^2 \times 5$.